

AtliQ Supply Chain & Finance Analytics

AtliQ operates in the hardware manufacturing and distribution environment, selling products such as mouse, keyboards, laptops and other electronic devices through direct, retail and distributor channels. The project applies SQL-based finance and supply-chain analytics to connect sales, customer, product, pricing, cost, freight, deductions, inventory and forecast information into management-oriented insights.

The objective of this project is not only to retrieve data but to convert transactional information into decisions around demand planning, customer management, product profitability, discount control, freight efficiency and inventory management.

Purpose of the Project

- Build a relational SQL data environment covering customers, products, sales, forecasts, gross prices, manufacturing costs, freight and deductions.
- Use SQL joins, aggregations, subqueries, common table expressions, window functions, stored procedures, user-defined functions and triggers.
- Translate operational data into business analysis covering sales trends, customer segmentation, product performance, market expansion, profitability, discounts, freight, seasonality, loyalty, forecast accuracy, channels, geography and inventory.
- Provide actionable management recommendations in finance and supply-chain analytics.

Business Objectives

- Improve demand planning by comparing forecast quantities with actual sales.
- Identify high-volume customers and understand concentration of observed revenue.
- Evaluate product profitability through gross price versus manufacturing cost.
- Measure the financial impact of pre-invoice deductions.
- Assess freight-cost differences across markets.
- Identify inventory movement and potential slow-moving products.
- Understand customer loyalty using observed purchase frequency.
- Support product mix and market expansion decisions using available demand and sales evidence.
- Create a repeatable SQL framework for operational monitoring and future dashboarding.

Business Users / Stakeholders

Stakeholder	How the analysis supports decisions
Supply Chain / Demand Planning	Forecast accuracy, sales trends, seasonality and inventory movement.
Finance	Revenue, gross profit, discount impact, manufacturing cost and margin analysis.
Sales / Key Account Management	Customer volume, revenue contribution and loyalty patterns.
Procurement / Operations	Freight-cost comparisons and inventory movement.
Product Management	Product-level sales, product mix and profitability.
Senior Management	Cross-functional view of revenue, demand, margin, customer and operational risks.

Table 1: Business Users / Stakeholders

Dataset Overview

The SQL model uses nine core business tables covering customer, product, monthly sales, monthly forecast, freight cost, gross price, manufacturing cost, post-invoice deductions and pre-invoice deductions. Additional SQL objects include audit/outlier logging, functions, stored procedures and triggers used to demonstrate controls and automation. The fiscal-year logic follows the project specification that is from September to August.

Table Name	Purpose of the Table	Key Information Stored	Business Use
dim_customer	This is the Customer Dimension table. It contains master information about AtliQ's direct customers who purchase products from the company.	Customer code, customer name, platform, channel, market, sub-zone and region.	Used to analyze customer performance, customer segmentation, sales by channel, market and region, and customer-level demand.
dim_product	This is the Product Dimension table. It contains the master information of products sold by AtliQ.	Product code, division, segment, category, product name, variant and inventory information.	Used for product performance, product segmentation, product mix, inventory analysis and profitability analysis.
fact_sales_monthly	This is the main monthly sales transaction/fact table. It records the quantity of products actually sold to customers over time.	Date, fiscal year/month, customer code, product code and sold quantity.	Used to measure actual sales volume, sales trends, customer demand, product performance, seasonality and forecast variance.
fact_forecast_monthly	This table contains the monthly sales forecast prepared for products and customers.	Date/forecast period, customer code, product code and forecast quantity.	Used to compare planned demand against actual sales and identify over-forecasting or under-forecasting.
fact_gross_price	This table stores the gross/base selling price of products for different fiscal years.	Product code, fiscal year and gross price.	Used to calculate gross revenue, pricing trends and product-level financial performance.
fact_manufacturing_cost	This table contains the manufacturing cost associated with products across fiscal years.	Product code, fiscal year and manufacturing cost.	Used to calculate gross profit and gross profit margin and evaluate product profitability.
fact_pre_invoice_deductions	This table records the pre-invoice deductions/discounts provided to customers before the invoice is finalized.	Customer, product/fiscal-year information and pre-invoice deduction percentage.	Used to understand the impact of customer discounts on gross revenue and net invoice sales.

fact_post_invoice_deductions	This table contains deductions that occur after the invoice has been generated, such as promotional or other post-invoice deductions.	Customer/product-related deduction information and deduction amounts/percentages.	Used to analyze additional revenue deductions and their impact on realized sales value.
fact_freight_cost	This table stores information about freight/logistics costs associated with products or markets.	Market and freight/other cost percentages.	Used to compare logistics costs across markets and identify opportunities for supply-chain cost optimization.

Table 2: Dataset Overview

Methodologies

- Relational data modelling with dimension and fact tables.
- Primary-key and foreign-key controls for data integrity.
- SQL joins to combine transactional and master data.
- GROUP BY and aggregate functions for management KPIs.
- Common table expressions and subqueries for multi-step calculations.
- Window functions such as LAG and ranking functions for trend and comparative analysis.
- PIVOT for forecast-versus-actual presentation.
- Stored procedures and user-defined functions for reusable calculations.
- Triggers for audit, inventory and data-validation demonstrations.
- Variance analysis using forecast quantity versus actual quantity.
- Margin analysis using gross price less manufacturing cost.
- Customer segmentation using observed volume/revenue and purchase frequency.

Key Findings

Sales and Revenue

The observed dataset records 263 units of sales and gross revenue of 4,048.94.

The sales-trend output shows 263 units for September 2017 for product A0118150101, making this product the central observed sales driver in the supplied result set.

All observed channel revenue is attributed to Direct sales, and all observed sales volume is concentrated in India/APAC.

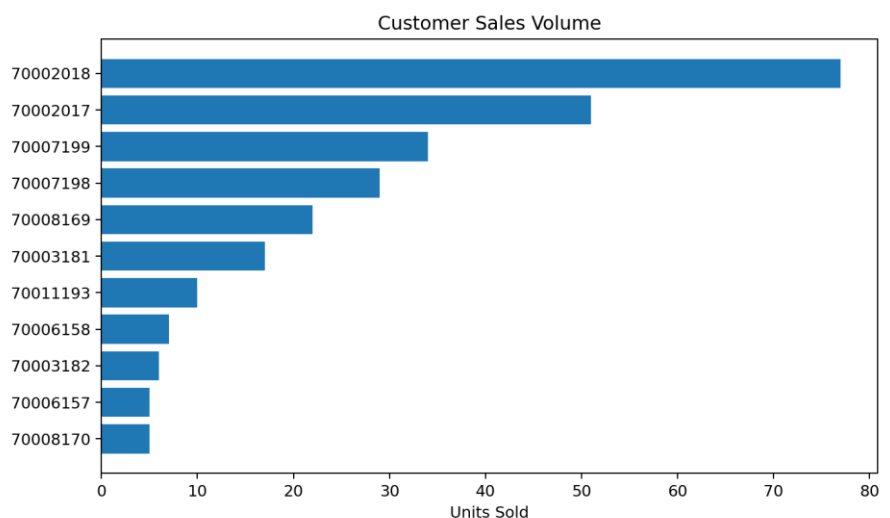


Diagram 1: Customer Sales Volume

Customer Performance

Customer 70002018 is the largest observed customer by both quantity (77 units) and gross revenue (1,185.43). The top customer represents approximately 29.3% of observed unit sales. The customer base therefore warrants concentration monitoring, although the dataset is too small and too concentrated to infer a full enterprise-wide customer concentration risk.

Customer	Units	Gross Revenue	Share of observed units
70002018	77	1,185.43	29.3%
70002017	51	785.16	19.4%
70007199	34	523.44	12.9%
70007198	29	446.46	11.0%
70008169	22	338.69	8.4%
70003181	17	261.72	6.5%

Table 3: Customer Performance

Profitability and Cost

The supplied cost analysis shows observed gross-profit margins between approximately 69.07% and 71.41% across the listed product-year observations. For A0118150101, the reported margin is 70.00% in 2018, 70.89% in 2019, 69.07% in 2020 and 71.05% in 2021.

This indicates a relatively stable margin band in the supplied records, while still showing year-to-year movement that should be monitored.

Product / Year	Manufacturing Cost	Gross Price	Gross Margin
A0118150103 / 2020	6.3264	22.1317	71.41%
A0118150103 / 2019	5.5306	19.3442	71.41%
A0118150102 / 2018	5.6036	19.5875	71.39%
A0118150102 / 2019	5.3235	18.5595	71.32%
A0118150102 / 2020	5.7180	19.8577	71.21%
A0118150101 / 2021	5.5172	19.0573	71.05%
A0118150101 / 2019	4.2033	14.4392	70.89%
A0118150102 / 2021	6.2835	21.4565	70.72%
A0118150101 / 2018	4.6190	15.3952	70.00%
A0118150103 / 2018	5.9469	19.3630	69.29%
A0118150101 / 2020	5.0207	16.2323	69.07%

Table 4: Profitability and Cost

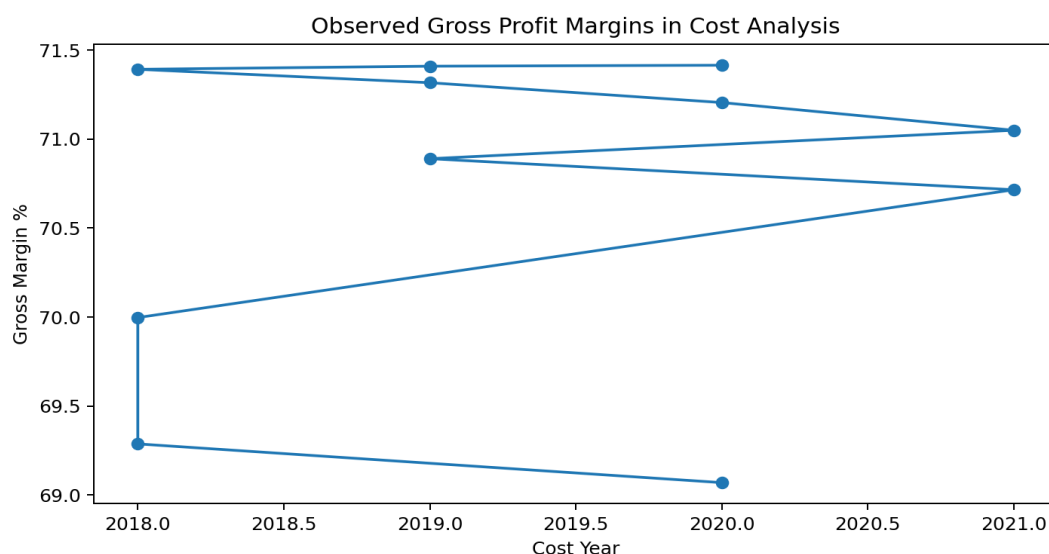


Diagram 2: Profitability and Cost

Discount Impact

Pre-invoice deductions materially reduce gross revenue. Like, customer 70002018 has a 29.56% pre-invoice discount, equivalent to 350.41 on gross revenue of 1,185.43, leaving 835.02 of net invoice revenue in the supplied calculation.

Customer	Pre-invoice discount	Gross revenue	Discount amount	Net invoice revenue
70002017	8.24%	785.16	64.70	720.46
70002018	29.56%	1,185.43	350.41	835.02
70003181	5.36%	261.72	14.03	247.69

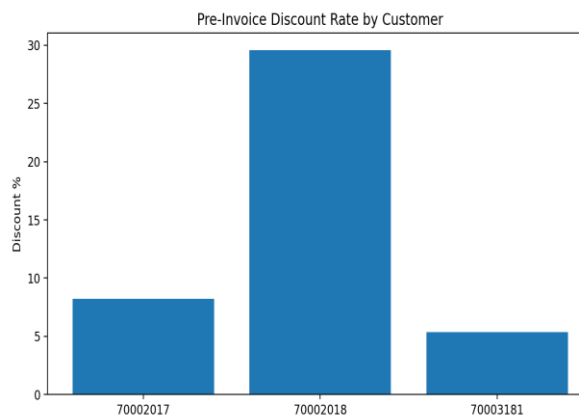


Diagram 3: Discount Impact

Table 5: Discount Impact

Forecast Accuracy and Demand Planning

The strongest operational finding is the gap between forecast and actual demand: forecast quantity is 81 units while actual sales are 263 units, a positive variance of 182 units. The project's formula reports actual/forecast $\times 100$, resulting in 324.69%.

This 324.69% figure should be described carefully: it is the project's actual-to-forecast ratio, not a conventional forecast-accuracy score. The practical business message is that actual demand was substantially above forecast and the planning process under-estimated demand.

Customer	Forecast	Actual	Variance	Actual/Forecast
70002017	18	51	33	283.30%
70002018	11	77	66	700.00%
70003181	9	17	8	188.90%
70003182	6	6	0	100.00%
70006157	5	5	0	100.00%
70006158	6	7	1	116.70%
70007198	4	29	25	725.00%
70007199	7	34	27	485.70%
70008169	2	22	20	1100.00%
70008170	8	5	-3	62.50%
70011193	5	10	5	200.00%

Table 6: Forecast Accuracy and Demand Planning

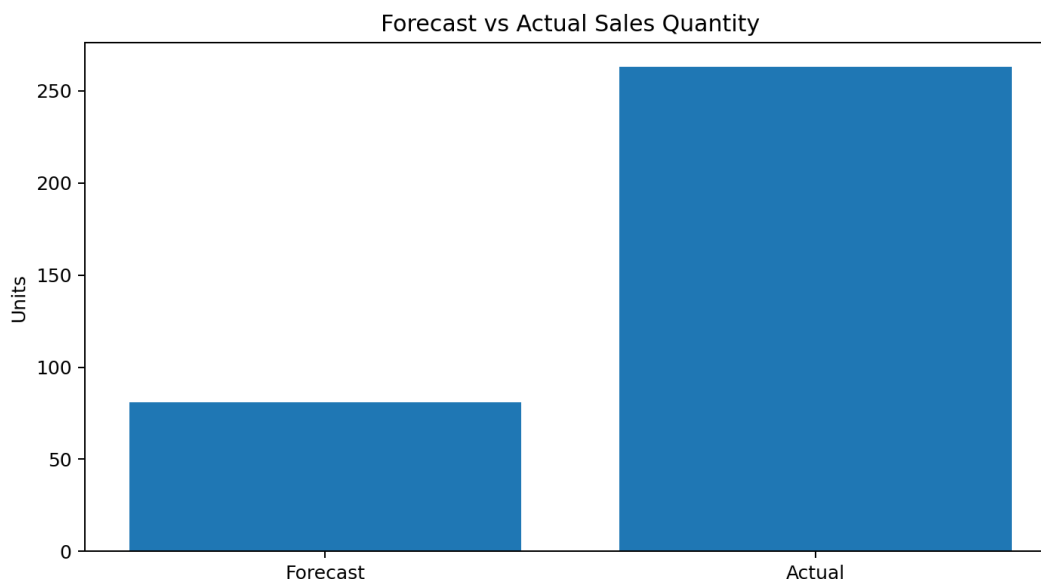


Diagram 4: Forecast Accuracy and Demand Planning

Freight Cost

India shows the highest average freight percentage among the three listed markets at 2.78%, compared with 2.26% in Germany and 1.91% in Indonesia.

Market	Avg Freight %	Avg Other Cost %	Min Freight %	Max Freight %
India	2.780%	0.340%	2.190%	3.090%
Germany	2.260%	0.600%	2.260%	2.260%
Indonesia	1.907%	0.390%	1.870%	1.950%

Table 7: Freight Cost

Because the underlying result set is limited, this should be treated as a comparative observation rather than proof that India is structurally inefficient.

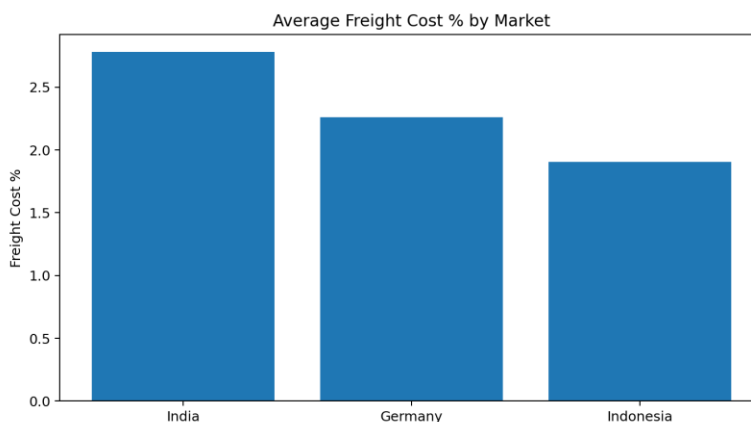


Diagram 5: Freight Cost

Customer Loyalty

All 11 observed active support customers are classified as Low Loyalty, with an average of one active purchase month and one fiscal year with purchases. This indicates limited repeat-purchase evidence in the supplied sample. It may also reflect the deliberately small/sample nature of the dataset.

Inventory

Within the support inventory records, A0118150101 has sales activity against an inventory count of 1,000, producing a simple turnover proxy of 0.263. A0118150102 and A0118150103 show no observed sales, so they are classified as slow/no movement. The measure is only a simple proxy because historical inventory snapshots are unavailable.

Product	Inventory	Sales Qty	Simple Turnover	Status
A0118150101	1000	263	0.263	Normal
A0118150102	1000	0	0	Slow / No Movement
A0118150103	1000	0	0	Slow / No Movement

Table 8: Inventory

Forecast by Customer – Key Variances

Customer-level forecast variance is highly uneven. Customer 70008169 has actual sales of 22 against a forecast of 2, while customer 70007198 has 29 against 4. Conversely, customer 70008170 has actual sales of 5 against a forecast of 8, making it the only observed customer in this block with actual demand below forecast.

Actionable Recommendations

- Demand planning: Recalibrate forecasts using customer-level actual demand and rolling history, as large positive variances indicate material under-forecasting in several customer-product combinations.
- Forecast governance: Track bias and absolute error separately from the project's actual/forecast ratio. It will prevent a ratio above 100% from being interpreted as 'high accuracy'.
- Customer management: Create targeted retention/reorder campaigns for high-volume customers and monitor repeat purchases as all observed customers are currently low-loyalty in the supplied sample.
- Inventory: Prioritize replenishment planning for A0118150101 and investigate inactive stock for A0118150102/A0118150103 because as observed demand is concentrated in one product.
- Discount governance: Review high pre-invoice discount customers, especially 70002018, against volume and margin contribution as discounts materially reduce net invoice revenue.
- Freight: Review logistics cost drivers for India and compare route/carrier/service-level economics as India has the highest listed average freight percentage.
- Product mix: Use margin plus sales volume together when deciding product focus as high margin alone is insufficient without evidence of demand.
- Data enhancement: Add acquisition spend, competitor prices, satisfaction scores and campaign data. These fields are required to calculate CAC, competitor benchmarking, satisfaction and campaign effectiveness.

Insights

- Demand planning is the clearest improvement opportunity because actual sales substantially exceed forecast in the observed period.
- The sales base is concentrated, one product accounts for the observed 263-unit sales result.
- Customer 70002018 is the largest observed account, but it also carries a comparatively high 29.56% pre-invoice discount.
- Observed product-year gross margins are consistently high and relatively stable around the 69–71% range.
- The inventory output suggests a need to distinguish productive inventory from non-moving stock rather than treating all products equally.
- The available dataset supports operational and financial diagnostics but does not support several requested strategic analyses without additional data.

Data Limitations

- The output includes technical/support records; these should not be presented as real-world customer identities or audited company figures.
- The observed result set is concentrated in one product, one channel and one market/region, limiting generalization.
- CAC cannot be calculated without acquisition spend and acquired-customer data.
- Competitor benchmarking cannot be calculated without competitor/product-price data.
- Customer satisfaction cannot be calculated without survey/feedback data.
- Marketing campaign effectiveness cannot be calculated without campaign, spend and response data.
- The inventory turnover output is a simple proxy because historical inventory levels are unavailable.

The project's forecast metric is an actual-to-forecast ratio; it should not be labelled as a standard accuracy percentage without clarification.

Conclusions

The SQL project demonstrates an end-to-end finance and supply-chain analytics workflow, moving from relational data preparation to customer, product, profitability, discount, freight, inventory and forecasting analysis. The supplied results point to a business environment where demand forecasting, discount governance and product/inventory focus are more immediately actionable than broad market conclusions.

For a production deployment, the next stage should be to expand the historical data population, introduce standardized forecast-error metrics, capture inventory snapshots, and add the missing datasets for customer acquisition, competitor pricing, customer satisfaction and marketing campaigns. These additions would allow the current SQL framework to evolve into a management dashboard and recurring decision-support system.